Madhusudhan Reddy Pittu

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Curriculum Vitae

Interests

I am broadly interested in Theoretical Computer Science. My primary areas of interest are Algorithms, Combinatorics, and Optimization. Most of my work is on Approximation algorithms for geometric problems.

Education

2021– 2025 **Ph.D. in Computer Science**, *Carnegie Mellon University*, Pittsburgh.
Advisor: Prof. Anupam Gupta, and Prof. David P. Woodruff

2017–2021 **B.Tech. in Computer Science and Engineering**, *Indian Institute of Technology Kharagpur*, India.
Advisor: Asst. Prof. Palash Dey

Publications

Max-Cut with Multiple Cardinality Constraints.
with Yury Makarychev, and Ali Vakilian
APPROX, 2025

Guessing Efficiently for Constrained Subspace Approximation.
with Aditya Bhaskara, Sepideh Mahabadi, Ali Vakilian, and David P. Woodruff
International Colloquium on Automata, Languages, and Programming (**ICALP**), 2025

Approximation Algorithms for the Weighted Nash Social Welfare via Convex and Non-Convex Programs.
with Adam Brown, Aditi Laddha and Mohit Singh
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), 2024

The Price of Explainability for Clustering.
with Anupam Gupta, Ola Svensson and Rachel Yuan
IEEE Symposium on Foundations of Computer Science (**FOCS**), 2023

Efficient Determinant Maximization for All Matroids.
with Adam Brown, Aditi Laddha and Mohit Singh
Arxiv, 2022

Determinant Maximization via Matroid Intersection Algorithms.
with Adam Brown, Aditi Laddha, Mohit Singh and Prasad Tetali.
IEEE Symposium on Foundations of Computer Science (**FOCS**), 2022

A 3-Approximation Algorithm for Maximum Independent Set of Rectangles.
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke and Andreas Wiese.
ACM-SIAM Symposium on Discrete Algorithms (**SODA**), 2022

A $(2+\varepsilon)$-Approximation Algorithm for Maximum Independent Set of Rectangles.
with Waldo Gálvez, Arindam Khan, Mathieu Mari, Tobias Mömke and Andreas Wiese.
Arxiv, 2021

On Guillotine Separability of Squares and Rectangles.
with Arindam Khan.
APPROX-RANDOM, 2020

Visits/Internships

Spring 2025 **Student Researcher**, *Google*.
Hosts: Sreenivas Gollapudi, Kostas Kollias

Summer 2023 **Non-Degree Visiting Student**, *Toyota Technological Institute at Chicago*.
Hosts: Ali Vakilian, Yury Makarychev, Julia Chuzhoy

Fall 2022 **Visiting Graduate Student**, *Simons Institute for the Theory of Computing*.
Data-Driven Decision Processes program

Summer 2020 **Visiting Student Research Internship (Remotely)**, *Georgia Tech*, Atlanta, USA.
Hosts: Prasad Tetali and Mohit Singh

Summer 2019 **Narendra Internship**, *Indian Institute of Science*, Bengaluru.
Host: Arindam Khan

Awards and Achievements

- **IITKGP Foundation-USA International Internship Award**, 2020
- Secured **Bronze Medal** representing India at the 57^{th} **International Mathematical Olympiad (IMO)**, 2016
- **Infosys award** for excellent performance in International Olympiads, 2016
- **Best Solution Award** in IMO training camp 2015

Talks/Presentations

Max-Cut with Multiple Cardinality Constraints.
APPROX, 2025

Guessing Efficiently for Constrained Subspace Approximation.
ICALP, 2025

Efficient Determinant Maximization for All Matroids.
IISc-MSR seminar at CSA department IISc, Bengaluru, 2022

A 3-Approximation Algorithm for Maximum Independent Set of Rectangles.
Symposium on Discrete Algorithms (**SODA**) virtual conference, 2022

On Guillotine Separability of Squares and Rectangles.
Highlights of Algorithms (**HALG**) virtual conference, 2020

On Guillotine Separability of Squares and Rectangles.
APPROX-RANDOM virtual conference, 2020